VEKTORSKA ANALIZA

65. – 88.

1. dio

Zad. 66.) Parametrizirajte krivulju $x^2+(y+1)^2=1$ orijentiranu u smjeru suprotnom od gibanja kazaljke na satu.

$x=\cos(t)$

$y=\sin(t)$

odnosno ovo je translatirana kružnica polumjera 1

$x-x_0 = r\cos(t) \longrightarrow$

$y-y_0 = r\sin(t) \longrightarrow$ ($y_0 \to -1$)

$$\boxed{\begin{aligned} x &= \sin(t) \\ y &= r\sin(t)-1 \end{aligned} \quad , t\in[0,2\pi]}$$

Zad. 67.) Parametrizirajte krivulju $x^2+6x+9y^2+54y=-81$

$$\underbrace{x^2+6x+9}\ -9+\underbrace{9y^2+54y+81}\ \cancel{-81}=\cancel{-81}$$

$$(x+3)^2+9(y+3)^2=9 \ /:9 \qquad \longrightarrow \text{elipsa}$$

$$\frac{(x+3)}{3^2}+\frac{(y+3)^2}{1}=1$$

elipsini parametri:

$x=a\cdot r\cdot\cos\varphi$

$y=b\cdot r\cdot\sin\varphi$

$a=3$, $b=1$

$$\Longrightarrow \boxed{\begin{aligned} x &= 3\cos t-3 \\ y &= \sin(t)-3 \end{aligned}}$$

#4

Zadatak 68.) Krivulja je zadana kao presjek ploha $z=-8(x^2+y^2)$ i $z=-8(x+y)$. Tangenta na tu krivulju u točki $T\left(\frac{1-\sqrt{2}}{2}, \frac{1}{2}, z_T\right)$ ima parametarsku jednadžbu:

* Jednadžba pravca prosz točku $T_1(x_1,y_1,z_1)$ zadanog vektora smjera $s=s_1\vec{i}+s_2\vec{j}+s_3\vec{k}$

$$\hookrightarrow \frac{x-x_1}{s_1}=\frac{y-y_1}{s_2}=\frac{z-z_1}{s_3}$$

$\Longrightarrow$ parametrizirano:

$$\begin{cases} x(t)=x_1+t\cdot s_1 \\ y(t)=y_1+t\cdot s_2 \\ z(t)=z_1+t\cdot s_3 \end{cases}$$

* <u>Jednadžba tangente na krivulju</u> $r(t)=x(t)\vec{i}+y(t)\vec{j}+z(t)\vec{k}$

$S(t_0)=(x'(t_0), y'(t_0), z'(t_0))$

$x(u)=x_0+u\cdot x'(t)$

$y(u)=y_0+u\cdot y'(t)$

$z(u)=z_0+u\cdot z'(t)$

krivulja je: $x^2-x+y^2-y=0$

koristim implicitnu derivaciju:

$$\frac{\partial f}{\partial y}=y'=-\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}=-\frac{2x-1}{2y-1}=\text{nagib}$$

$$z_0=-8\cdot\left(\frac{1-\sqrt{2}}{2}\right)-\frac{8}{2}$$

$$=-4(1-\sqrt{2})-4=-4(2-\sqrt{2})$$

oblik pravca (tangente)

$$y-y_0=\frac{-2x+1}{2y-1}(x-x_0)$$

$$y-\frac{1}{2}=\frac{-2\cdot\frac{1}{2}+\sqrt{2}+1}{1-1}\left(x-\frac{1}{2}+\frac{\sqrt{2}}{2}\right)$$

Zad. 69.) $z = 2(x^2+y^2)$ $\quad z = 2(x+y)$, $T\left(\frac{1-\sqrt{2}}{2}, \frac{1}{2}, z_T\right)$

Parametarska jednadžba tangente krivulje:

Zad 70.) Usmjerena derivacija vekt polja u smjeru nekog vektora je

(B) Vektorsko polje

Zadatak 71.) Odredi usmjerenu deriv. polja $r^4\vec{r}$ po vektoru $s = \frac{i+k}{\sqrt{2}}$ u točki $T(1,0,1)$, ako je $\vec{r}$ radij vektor točke, a r njegov iznos.

* Usmjerena DERIV. (MAT2): $\frac{\partial f}{\partial \vec{v}}(T_0) = \nabla f \cdot \vec{v_0} = \frac{\partial f}{\partial x}(x_0,y_0)v_{0_1} + \frac{\partial f}{\partial y}(x_0,y_0)v_{0_2}$

$F(\vec{r}) = r^4\vec{r}$, $\quad r = \sqrt{x^2+y^2+z^2}$, $\quad \vec{r} = (x,y,z)$

$F(x,y,z) = r^4 \cdot (x,y,z) = (x^2+y^2+z^2)^2\,(x,y,z)$

$\nabla F \cdot s = (s\cdot\nabla F_x,\ s\cdot\nabla F_y,\ s\cdot\nabla F_z)\big|_T$

$$\frac{\partial F_x}{\partial x} = \left((x^2+y^2+z^2)^2 \cdot x\right)' = (x^2+y^2+z^2)^2 + x\cdot 2\cdot 2x\,(x^2+y^2+z^2)$$
$$= r^4 + 4x^2\cdot r^2$$

$$\frac{\partial F_x}{\partial y} = \left(x(x^2+y^2+z^2)^2\right)' = x\cdot 2\cdot 2y\,(x^2+y^2+z^2) = 4xy\cdot r^2$$

$$\frac{\partial F_x}{\partial z} = \left(x(x^2+y^2+z^2)^2\right) = x\cdot 2\,(x^2+y^2+z^2)\cdot 2z = 4xz\cdot r^2$$

∇F_x

$$\frac{\partial F_y}{\partial y} = \left(y\,(x^2+y^2+z^2)^2\right)' = (x^2+y^2+z^2)^2 + y\cdot 2\,(x^2+y^2+z^2)\cdot 2y = r^4 + 4y^2r^2$$

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y} = 4xyr^2 \qquad \frac{\partial F_y}{\partial z} = y\cdot 2\cdot 2z\cdot r^2 = 4yzr^2$$

$$\frac{\partial F_z}{\partial z} = r^4 + 4z^2r^2 \qquad \frac{\partial F_z}{\partial x} = 4xzr^2 \qquad \frac{\partial F_z}{\partial y} = 4yzr^2$$

$s = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$, $\quad r\big|_T = \sqrt{1+0+1} = \sqrt{2}$

$s\cdot\nabla F_x = (4 + 4\cdot 1\cdot 2)\frac{1}{\sqrt{2}} + 0 + 4\cdot 1\cdot 2\cdot\frac{1}{\sqrt{2}} = 10\sqrt{2}\,i$

$s\cdot\nabla F_y = \emptyset \qquad = \emptyset\, j$

$s\cdot\nabla F_z = 4\cdot 2\cdot\frac{1}{\sqrt{2}} + \emptyset + (4+4\cdot 2)\frac{1}{\sqrt{2}} = 10\sqrt{2}\,k$

$$\Rightarrow \boxed{(\nabla F\cdot s) = 10\sqrt{2}\,(\vec{i}+\vec{k})}$$

Zadatak 72) Usmjerena derivacija polja $r^6 \vec{r}$ po vektoru

$s = \frac{i + j + k}{\sqrt{3}}$, $T(1,1,-1)$ ako je $\vec{r}$ radij-vektor točke, a r njegov iznos

$$F(\vec{r}) = r^6 \vec{r} \longrightarrow F(x,y,z) = (x^2+y^2+z^2)^3 \cdot (x, y, z)$$

$$\frac{\partial F_x}{\partial x} = \left(x(x^2+y^2+z^2)^3\right)' = (x^2+y^2+z^2)^3 + x \cdot 3 \cdot 2x\,(x^2+y^2+z^2)^2$$

$$= \underbrace{r^6 + 6x^2 \cdot r^4}$$

$$\frac{\partial F_x}{\partial y} = x \cdot 3 \cdot 2y\,(x^2+y^2+z^2)^2 = \underbrace{6xy \cdot r^4}$$

$$\frac{\partial F_x}{\partial z} = x \cdot 3 \cdot 2 \cdot z\,(x^2+y^2+z^2)^2 = \underbrace{6xz\,r^4}$$

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y} = \underbrace{6xyr^4} \qquad \frac{\partial F_y}{\partial y} = \underbrace{r^6 + 6 \cdot y^2 r^4} \qquad \frac{\partial F_y}{\partial z} = \underbrace{6 \cdot yzr^4}$$

$$\frac{\partial F_z}{\partial x} = \underbrace{6xzr^4} \qquad \frac{\partial F_z}{\partial y} = \underbrace{6yzr^4} \qquad \frac{\partial F_z}{\partial z} = \underbrace{r^6 + 6z^2r^4}$$

$$s = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad r = \sqrt{3} \quad ; \quad r^4 = 9, \quad r^6 = 27$$

$$\nabla F \cdot s = (\nabla F_x \cdot s, \ \nabla F_y \cdot s, \ \nabla F_z \cdot s) =$$

$$(27 + 6 \cdot 9) + \cancel{6 \cdot 9} - \cancel{6 \cdot 9} = 27\sqrt{3} \quad \ldots \text{ i tako } y$$

$$z \rightarrow -9\sqrt{3} \quad \Longrightarrow \quad \boxed{27\sqrt{3}\,i + 27\sqrt{3}\,j - 9\sqrt{3}\,k}$$

Zadatak 73.) $r = \|\vec{r}\|$, $f(r) = r^{17} \cdot \ln(r)$, $s = \frac{i+j}{\sqrt{2}}$

u točki $T\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$$\cancel{\frac{\partial(\ln(r))\vec{r}}{\partial \vec{s}} = (\vec{s}\cdot\nabla)(\ln(r)\cdot\vec{r}) = ((\vec{s}\cdot\nabla)\ln r)\cdot\vec{r} + \ln(r)((\vec{s}\cdot\nabla)\vec{r})}$$

$$\cancel{= \left(\frac{i+j}{\sqrt{2}} \cdot \frac{1}{r}\cdot\vec{r}_0\right)\vec{r} + \ln(r)\cdot\frac{i+j}{\sqrt{2}}}$$

↳ primjer iz skripte

* u točki $T\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ vrijedi $\vec{r} = \frac{1}{\sqrt{3}}i + \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k$

i $\underline{|r| = 1}$

$f = r = \|r\|$

↳ $\nabla f = f'(r)\cdot\nabla r \longrightarrow f'(r) = 17\cdot r^6\cdot\ln(r) + r^{\cancel{17}}\cdot\frac{1}{\cancel{r}}$

$\nabla r = \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right)$

$\underline{f'(r) = 17\cdot r^6 \ln(r) + r^{16}}$

$$\nabla f = (17\cdot r^6\ln(r) + r^{16})\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right) = (17\ln(r)+1)\, r^{16}\cdot\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right)$$

$$\nabla f = (17\ln(r)+1)\cdot r^{15}\cdot(x, y, z)$$

u točki $T \longrightarrow r = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \Rightarrow r = 1 \Rightarrow \ln(r) = 0$

$$\nabla f_T = (1\cdot 1^{15})(x, y, z) = \underline{\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)}$$

$$\nabla f_T \cdot s = \left(\frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{2}}, 0\right) = \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}} = \frac{\sqrt{6}\cdot\cancel{2}}{\cancel{6}} = \boxed{\frac{\sqrt{6}}{3}}$$

Zad. 74) Usmjerena derivacija polja ovisnog o $r = \|\vec{r}\|$

Zadanog s $f(r) = r^{11} \ln(r)$ u smjeru vektora $s = \frac{i+j}{\sqrt{2}}$, $T\left(\frac{1}{\sqrt{3}} \dots\right)$

$$\nabla f = f'(r) \cdot \nabla r \qquad f'(r) = 11 \cdot r^{10} \cdot \ln(r) + r^{10}$$

$$\nabla r = \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right)$$

$$\nabla f = (11 \ln(r) + 1) \cdot r^{10} \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right) = (11 \cdot \ln(r) + 1) \cdot r^{9} (x, y, z)$$

$$\|r\|\Big|_T = \sqrt{\frac{1}{3} \times 3} = 1 \rightarrow r = 1 \qquad \nabla f\Big|_T = 1 \cdot 1 \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$(\nabla f \cdot s) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}} = \frac{\sqrt{6}}{3}$$

Zadatak 75.) $f(x,y,z) = \underbrace{(yz + ye^{xy} + 2x)}_{v_1} i + \underbrace{(xz + xe^{xy})}_{v_2} j + \underbrace{(axy + z^{13})}_{v_3} k$

Odredite sve moguće vrijednosti parametra $a \in \mathbb{R}$ tako da polje bude potencijalno

Da bi f bilo potencijalno polje v_1, v_2 i v_3 trebaju biti gradijenti neke skalarne funkcije $p(x,y,z)$.

$\Rightarrow$ Da bi bilo vekt. polje potenc mora vrijediti $\operatorname{rot} \vec{v} = \emptyset \rightarrow \nabla \times \vec{v} = \emptyset$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz + ye^{xy} + 2x & xz + xe^{xy} & axy + z^{13} \end{vmatrix} = i\left(\frac{\partial}{\partial y}(axy + z^{13}) - \frac{\partial}{\partial z}(xz + xe^{xy})\right)$$

$$- j\left(\frac{\partial}{\partial x}(axy + z^{13}) - \frac{\partial}{\partial z}(yz + ye^{xy} + 2x)\right)$$

$$+ k\left(\frac{\partial}{\partial x}(xz + xe^{xy}) - \frac{\partial}{\partial y}(yz + ye^{xy} + 2x)\right)$$

$$\operatorname{rot} \vec{v} = i(ax - x) - j(ay - y) + k\left(\cancel{z + e^{xy} + xye^{xy} - (z + e^{xy} + xye^{xy})}\right)$$

pokrati se

$$\operatorname{rot} \vec{v} = 0 \longrightarrow \begin{matrix} x(a-1) = 0 \\ -y(a-1) = 0 \end{matrix} \longrightarrow a = 1$$

DK

Zadatak 76.) polje treba vih potencijalo za sve a:

$$f(x,y,z) = (yz + ye^{xy} + 2x)i + (xz + xe^{xy})j + (axy + z^2)k$$

uvjet $\Rightarrow$ rot $\vec{v} = \emptyset \longrightarrow$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz + ye^{xy} + 2x & xz + xe^{xy} & axy + z^2 \end{vmatrix}$$

$$= i\left(\frac{\partial}{\partial y}(axy + z^2) - \frac{\partial}{\partial z}(xz + xe^{xy})\right) - j\left(\frac{\partial}{\partial x}(axy + z^2) - \frac{\partial}{\partial z}(yz + ye^{xy} + 2x)\right)$$

$$+ k\left(\frac{\partial}{\partial x}(xz + xe^{xy}) - \frac{\partial}{\partial y}(yz + ye^{xy} + 2x)\right)$$

$$= i(ax - x) - j(ay - y) + k(z + e^{xy} + xye^{xy} - z - e^{xy} - xye^{xy}) = \text{isto}$$

$a = 1$

Zadatak 77.)

Izračunajte $\nabla(\vec{r} \times \vec{a}) + \nabla r$ tako da $\vec{r}$ radij vektor točke, r njegov iznos, a $\vec{a}$ konstantan vektor.

$$\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right)$$

Laplaceov operator

divergencija $(\vec{r} \times \vec{a}) = \nabla(\vec{r} \times \vec{a}) = \nabla(\vec{r} \times \underline{a}) + \nabla(\underline{\vec{r}} \times a)$

rot od r je $\emptyset$!

$$= \vec{a}(\nabla \times \vec{r}) + \vec{r}(\vec{a} \times \nabla) = \underbrace{\vec{a}(\nabla \times \vec{r})}_{\emptyset} - \underbrace{\vec{r}(\nabla \times \vec{a})}_{\emptyset}$$

$\nabla \times (\text{konst. vektor}) = \emptyset$

$\longrightarrow$ ostaje samo $\nabla r = \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right) = \frac{\vec{r}}{|r|} = \hat{r}_0$

Zadatak 78.) $\nabla\left(\frac{r^{28}\cdot\vec{r}}{a\cdot\vec{r}}\right)$

PRAVILO ZA DERIV KOLIČNIKA: $\nabla\left(\frac{f(\vec{r})\cdot\vec{r}}{\vec{a}\cdot\vec{r}}\right) = \frac{\nabla(r^{28}\cdot\vec{r})(\vec{a}\cdot\vec{r}) - r^{28}\cdot\vec{r}\,\underline{\underline{\nabla(a\cdot r)}}}{(\vec{a}\cdot\vec{r})^2}$ $\uparrow a$

$$= \frac{\nabla(r^{28}\cdot\vec{r})(\vec{a}\cdot\vec{r}) - r^{28}\cdot\vec{r}\cdot\vec{a}}{(\vec{a}\cdot\vec{r})^2}$$

$$\nabla(r^n\cdot\vec{r}) = r^n\underbrace{\nabla\vec{r}}_{1} + \vec{r}\,\nabla r^n$$

$$\nabla(r^{28}\cdot\vec{r}) = r^{28}\underbrace{\nabla\vec{r}}_{I} + \vec{r}\cdot\nabla r^{28} = r^{28}\cdot I + \vec{r}\,28\cdot r^{26}\cdot\vec{r} = r^{28}I + 28\cdot r^{26}\underline{\underline{\vec{r}\times\vec{r}}}$$

$\rightsquigarrow$ ali postoji već poznata formula koju možemo primjeniti:

$$\nabla\left(\frac{r^n\vec{r}}{\vec{a}\cdot\vec{r}}\right) = \frac{r^n}{\vec{a}\cdot\vec{r}}\left(I + n\frac{\vec{r}\times\vec{r}}{r^2} - \frac{\vec{r}\times\vec{a}}{\vec{a}\cdot\vec{r}}\right)\ldots$$ bla bla

nc zna

Zadatak 79.) $r = \|\vec{r}\|$, $\vec{r}$ je radij vektor, Δr^{15} iznosi:

$$\frac{\nabla r^n}{\vec{r}\cdot r^{n-2}} = n\cdot(n+1)\cdot r^{n-2} \rightarrow \nabla r^{15} = \boxed{15\cdot 16\cdot r^{13}}$$ C

Zadatak 80.) Δr^{19}, ?

$$19\cdot 20\, r^{17} \Longrightarrow 380 r^{17}$$

Zadatak 81.) Nađite vektor u čijem smjeru iz točke $T(1,-1)$ funkcija $f(x,y) = 16 - 6x^2 - 9y^2$ najbrže raste?

f najbrže raste u smjeru ∇f

↳ iznos je $|\nabla f(T_0)|$

$$\nabla f\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (-12x, -18y)$$

$$\nabla f(T_0) = (-12, 18) \Longrightarrow 3\nabla f(T_0) = -36i + 54j$$ A

Zad. 82.) Najbrži rast funkcije $f(x,y) = 14 - 7x^2 - 6y^2$, $T(1,-1)$

$$\nabla f(-14x, -12y) \longrightarrow -14, 12 \cdot \frac{7}{2} \longrightarrow \boxed{-49i + 42j}$$ D

Zad. 83.) Integral $\int_\Gamma \frac{-ds}{\sqrt{1+\frac{y^2}{4}}}$ po krivulji Γ koja je presjek plohe $x^2 - 2x + y^2 = 3$ i $z = x$ iznosi:

$z = x = t \qquad t^2 - 2t + y^2 = 0 \longrightarrow y^2 = 2t - t^2$, $\quad ds = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}$

$y = \pm\sqrt{t(2-t)}$

$$ds = \sqrt{1^2 + \left(\frac{1}{2\sqrt{t(2-t)}}(2t-2)\right)^2} = \sqrt{1 + \frac{\cancel{4}(t-2)^2}{\cancel{4}t\cancel{(t-2)}}} = \sqrt{1 - \frac{t-2}{t}}$$

$x^2 - 2x + 1 - 1 + y^2 = 3 \longrightarrow (x-1)^2 + y^2 = 4$

$x - 1 = 2\cos t \rightarrow z = x = 2\cos t + 1 \longrightarrow x' = -2\sin(t)$ $\qquad t \in [0, 2\pi]$

$y = 2\sin t \qquad y' = 2\cos(t)$

$$ds = \sqrt{4\sin^2(t) + 4\cos^2(t) + 4\sin^2(t)} = \sqrt{4 + 4\sin^2(t)}$$

$$y^2 = 4\sin^2(t) \longrightarrow \int_0^{2\pi} \frac{-2\cancel{\sqrt{1+\sin^2(t)}}}{\cancel{\sqrt{1+\sin^2(t)}}} = -2 \cdot 2\pi = \boxed{-4\pi}$$

Zad 84.) $\int_\Gamma \frac{-2ds}{5\sqrt{1+\frac{y^2}{4}}}$, $x^2 - 2x + y^2 = 3$, $z = x$

$ds = \dots = 2\sqrt{1+\sin^2(t)}$, $\quad y^2 = \dots = 4\sin^2(t)$

$$\int_0^{2\pi} \frac{-4\cancel{\sqrt{1+\sin^2(t)}}}{5\cancel{\sqrt{1+\sin^2(t)}}} = -\frac{4}{5} \cdot 2\pi = \frac{-8\pi}{5}$$ B

Zadatak 85.) Izračunajte $\int_C (2x^2+y^2)\,ds$ gdje je krivulja C presječnica plohe $x^2+2y^2=4$ i $z=y$ uz uvjet $y \geq 1$.

$$\frac{x^2}{4}+\frac{y^2}{2}=1 \qquad a=2 \qquad b=\sqrt{2}$$

$$x=2\cos(t) \qquad y=\sqrt{2}\cdot\sin(t)=z \qquad t\in[0,2\pi]$$

$$x'=-2\sin(t)$$

$$y'=z'=\sqrt{2}\cos(t)$$

$$ds=\sqrt{4\sin^2(t)+2\cos^2(t)+2\cos^2(t)}=\underline{\underline{2}}$$

$y \geq 1$

$$\vec{r}=(2\cos(t),\ \sqrt{2}\sin(t),\ \sqrt{2}\sin(t)) \qquad y=\sqrt{2}\sin(t)\geq 1 \longrightarrow t\in\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$$

$$2x^2+y^2 \longrightarrow 2\cdot 4\cos^2(t)+2\sin^2(t) \longrightarrow 2\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 8\cos^2(t)+2\sin^2(t)\,dt$$

$-1+8\cos^2(t)$

$$\cos 2t=\cos^2(t)-\sin^2(t)$$

$$\cos^2(t)=\frac{\cos 2t+1}{2} \qquad \sin^2(t)=\frac{1-\cos 2t}{2}$$

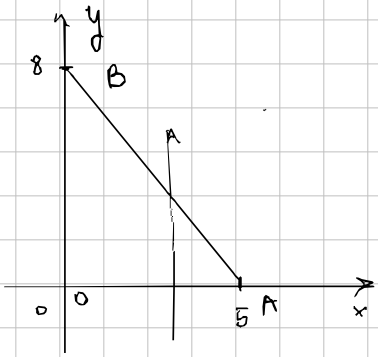
$$2\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 4\cos 2t+4+1-\cos 2t\,dt$$

$$2\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 5+3\cos 2t\,dt = 2\left(5\cdot\frac{\pi}{2}+3\cdot\frac{1}{2}\sin(2t)\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\right)=5\pi+3\cdot(-2)$$

$$\Rightarrow \boxed{5\pi-6}$$

86 $\int_\Gamma \left(\frac{x}{5}+\frac{y}{8}\right) ds$ Γ je rub trokuta s vrhovima u točkama

$A(5,0)$, $B(0,8)$, $O(0,0)$



$y = ax+b$

$8 = b$ $\quad 0 = a\cdot 5+8$

$y=0$ $\quad a = \frac{-8}{5} \rightarrow y = \frac{-8}{5}x+8$

$$\int_0^5 dx \int_0^{\frac{-8}{5}x+8} \left(\frac{x}{5}+\frac{y}{8}\right)\cdot ds \quad dy \qquad \sqrt{\frac{1}{25}+\frac{1}{64}} = \frac{\sqrt{89}}{40}$$

→ u ovom zadatku računamo luk trokuta a ne površinu → nije dvostruki

→ integracija po 3 dužine

1) $\{D-A\}$ → horizontalna $\quad y=0, \quad x=t \quad t\in[0,5]$

$ds = \sqrt{x'+y'} = 1 \rightarrow \frac{x}{5}+\frac{y}{8} = \frac{t}{5}+0 = \frac{t}{5}$

$$\int_0^5 \frac{t}{5}\cdot 1\,dt = \frac{1}{5}\cdot\frac{1}{2}t^2\Big|_0^5 = \boxed{\frac{25}{10} = \frac{5}{2}}$$

2) $\{B-O\}$ → vertikalna $\quad y=t \quad x=0 \rightarrow t\in[0,8]$

$ds = \sqrt{1} = 1 \qquad 0+\frac{t}{8} = \frac{t}{8} \qquad \frac{1}{8}\int_0^8 t\,dt = \frac{1}{16}t^2\Big|_0^8 = \frac{64}{16} = \boxed{4}$

3) $\{B-A\}$ $\quad y = \frac{-8}{5}x+8 \qquad y = -8\left(\frac{1}{5}x-1\right)$

$\rightarrow ds = \sqrt{8^2+5^2} = \sqrt{89} \qquad \frac{y}{8} = 1-\frac{1}{5}x = t$

$y = 8t, \quad 5(1-t) = x$

$t\in[0,1)$ jer

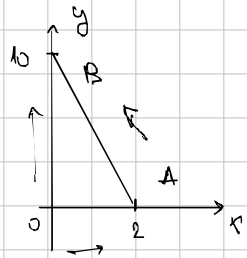
$t=0 \rightarrow (5,0)$, $t=1 \rightarrow (0,8)$

$$\int_0^1 \left(\frac{x}{5}+\frac{y}{8}\right)\sqrt{89}\,dt = \sqrt{89}\int_0^1 1-\cancel{t}+\cancel{t}$$

$= \sqrt{89}$

$\frac{5}{2}+4+\sqrt{89} \rightarrow \boxed{\sqrt{89}+\frac{13}{2}}$ A

Zadatak 87.) $\int_\Gamma \left(\frac{x}{2}+\frac{y}{10}\right) ds$ tocke: $\overset{O}{(0,0)}$, $\overset{A}{(2,0)}$, $\overset{B}{(0,10)}$



1) $(O-A)$ $\quad x=t, \quad y=0 \rightarrow ds = \underline{\underline{1}}$

$$\int_0^2 \frac{t}{2}\,dt = \frac{1}{4}t^2\Big|_0^2 = \underline{\underline{1}}$$

2) $\overline{OB}$ $\quad x=0, \quad y=t \rightarrow ds=1 \Rightarrow \int_0^{10} \frac{t}{10}\,ds = \frac{1}{\cancel{20}}\cdot\cancel{100} = \underline{\underline{5}}$

3) $\overline{BA}$ $\quad -\frac{10}{2}x+10 = y \quad /\cdot 10$

$$1-\frac{1}{2}x = \frac{y}{10} \longrightarrow \underline{y=10t}$$

$$2-x = 2t \longrightarrow \underline{x=2(1-t)}$$

$$ds = \sqrt{100+4} = \underline{\underline{\sqrt{104}}}$$

$$\int_0^1 (1-\cancel{t}+\cancel{t})\sqrt{104}\,dt = \underline{\underline{\sqrt{104}}}$$

Ukupno $\boxed{\Sigma = 6+\sqrt{104}}$ B

Zadatak 88.) $\int_\Gamma \left(\frac{x}{2}+\frac{y}{9}\right) ds$ $\quad A(2,0)$, $B(0,9)$, $O(0,0)$

$$\int_0^2 \frac{t}{2}\,dt = \underline{\underline{1}} \qquad \int_0^9 \frac{t}{9} = \frac{1}{18}\cdot 81 = \underline{\underline{\frac{9}{2}}}$$

$\underline{y=9t}$ $\qquad \underline{x=2(1-t)}$ $\qquad ds = \sqrt{81+4} = \sqrt{85}$

$$\sqrt{85}\int_0^1 1-t+t\,dt = \underline{\underline{\sqrt{85}}} \longrightarrow \boxed{\underline{\underline{\frac{11}{2}+\sqrt{85}}}}$$ A